

Intelligent IOT-based BMS for Integrated Solar with AC grid Fast Charging Systems in Electric Vehicles

1st Shreyas Thombare

MS student, Advanced Technology
Development Centre, IIT Kharagpur.
thombaress2017@gmail.com

2nd Pankaj Kumar

Assistant professor, Electrical
department, Dr.D.Y.Patil Institute of
Technology Pimpri Pune, India
pankaj.kumar@dypvp.edu.in

3rd Rajashree Bhokare

Assistant professor, Electrical
department, Dr.D.Y.Patil Institute of
Technology Pimpri Pune, India
rajashree.bhokre@dypvp.edu.in

4th Sachin Jadhav

Assistant professor, Electrical
department, Dr.D.Y.Patil Institute of
Technology Pimpri Pune, India
Sachinjadhav958@gmail.com

Abstract— Environmental pollution is the biggest concern these days' electric cars are introduced as a less polluting solution, especially for fossil fuels which are reduced considerably and there is no release of any harmful gas. Due to the long charging time of the battery, electric vehicles often face problems. This research paper aims to offer a portable charger coupled converter (AC-DC) with a solar charge controlled (DC-DC) converter used to charge lithium-ion battery packs used in EV technology. Intelligence added for the maintenance of safety standards through IoT-based BMS controlling, which helps for wide range data storage over the cloud and helps in data analysis & monitoring the important safety parameters of the battery along with accurate SoC (State of charge) & SoH (State of Health) estimation of any type of Li-ion battery chemistry cells. Most EV batteries catch fire it's hard to predict the reasons as all data will be lost with BMS hardware damage with temperature but it can be avoided by IOT-based data collection applicable to any type of battery used in EV. At the time when charging is full, the system automatically disconnects the charger from the battery. When this system is used, the time taken for charging is reduced and other parameters are monitored as well, and this is done under a highly specialized system of intelligence and related settings. It is important to note that the system of battery management has no impact on thermal performance.

Keywords— Battery Thermal Stabilization, Battery Parameter Monitoring, Internet of things, SoC & SoH estimation, Fast charging.

I. INTRODUCTION

The most recent trend across the globe these days includes electric vehicles. Electrical vehicles are replacing, the most conventional vehicle since their availability is limited and dependent on the presence of fossil fuels on Earth. When electrical devices are used, the biggest issue is the battery's operational temperature stability and the time taken by the battery to get charged. [1] In the charging conventional system, a single standard charger is used to charge the entire battery. A longer time is needed by it to charge the battery completely to its fuller rating. There is often a proposed charging system solution in these vehicles; where the connection of series & parallel number of cells makes a battery pack of electric car. The two parts of battery charging systems are synchronized at the same voltage and divided from both sides in parallel helping for higher rated charging current at the same time simultaneously. To increase and decrease the DC voltage, a bidirectional boost converter is used normally. On the other hand, at extreme levels of working cycles, it is supposed to work in either direction for the achievement of higher ratios up and down. On diodes and switches, this produces extra high stress and hence ends up in

the reversal of reset loss, higher EMI and lower efficiency relatively. With the help of this relay, the charger from the battery is connected and disconnected automatically. There are some battery parameters such as the status of battery charging, voltage, current, temperature etc., all these parameters are displayed on the displaying board. The tracking as well as monitoring is made easier by it. Therefore, when the basic tolerance limit is exceeded by any of these parameter, the BMS allows instant disconnection from the charging of the battery. The ways electric vehicles' charging time is reduced, deep discharge over temperature, and the ways to avoid overload are studied in this paper [2].

1.1 Necessity

Since there is a drastic increase in the global population these days, the demand for energy supply is rapidly increasing and hence in the future, it is expected that the world will face the scarcity of these resources. The available resources are even far lower and insufficient to fulfill the demands of the population in the future. Environmental pollution is the biggest concern these days; a lot of pollution is caused by the consumption of fuels and engines of vehicles that are injurious to human health. The use of electric vehicles is a modern need, and it is essentially a need of time to replace these vehicles with older and conventional motor vehicles. By the end of 2030, it has been even announced by the government that all conventional vehicles will have to be replaced by electrical vehicles. We all know that the current time requires electrical vehicles in the field of mobility, especially in the coming times. The time taken by the battery to charge is the main problem of electrical vehicles. The reduction of battery timings is essential for electric vehicles. The Proposed charging system is used to overcome this problem. The time taken to charge the battery is reduced considerably with the help of intelligent charging system. With the help of this system, as estimated, the time taken to charge the battery is reduced to half of the original time. The small and readily available engine sources are harnessed by the suggested system in this paper. In the coming times, the suggested research will be developed more as there are more and more advancements in security and safety [3].

Design an Electric Vehicle Battery Advanced Charging intelligent system is a result of the expansion of the battery limit, and, along these lines, it takes only half an hour to charge the 1-kWh battery with the super-fast charging rate of 1.4kW:

- The DC charger is outside the vehicle and along these lines is not liable for size or cost. In addition, such superchargers can interact with multiple-stage power-

sharing that allows for change of charge energy levels to suit the different battery chemistry.

- The improvement of battery performance by avoiding deep battery discharging and overcharging.
- To the analysis of the battery parameters of various kinds such as charging state, temperature current and voltage of the battery over cloud.
- Wide scale of data collection, long time recording & analysis in accurate levels helps in precision SoC & SoH calculations required for range estimation in electric vehicles.

1.2 Related Literature Work

In the modern-day transportation, electrical vehicles are used these days. One of the biggest concerns of using electrical vehicles is the time taken to charge the battery and this time is often long. A lot of investigations are still ongoing in this department to overcome this problem. Various techniques and systems of charging have been suggested by investigative workers for modern-day electric vehicles. In a few literature studies, it's observed that some kinds of suggested systems were published over fast charging in the past. In this research paper, all they are mentioned is to discover some modern technologies and their limitations, and this subject will be helpful in the research taking place in the future. A few technical terms are mentioned in this paper and are listed below in the bibliographic investigative survey. [4]. In the field of battery management, advanced techniques and control techniques to check and enhance the performance of the battery are always important under research. A plug-in electric vehicle charger with a High Step-Up and High-Efficiency Fuel Cell was suggested by a group of investigative workers named W. Li, L. Fan, Y. Zhao, X. He, D. Xu, and B. Wu, [5]. With the help of an active clamp Fly back-forward converter, the performance of vehicles and optimization of components is done. Supercapacitor charging system for electric vehicles was demonstrated by several investigative workers named K. Hatua, S. Dutta, A. Tripathi, S. Baek, G. Karimi and S. Bhattacharya, [6]. The work has been done in this system to charge the battery cells with the new substitution design of transformer less intelligence power substation. A high step-up current fed multi-resonant converter with output voltage doubler was suggested by some investigative workers named D. Li, Bo Liu, Bo Yuan, Xu Yang, J. Duan, and J. Zhai [7]. They suggested an output voltage doubler with a high step-up current fed with multiple resonant converters. The future aspect of electric vehicles and their fast charging was suggested by research worker named Zhang. C, Jiang. J, Gao. Y, Zhang. W, Liu. Q, Hu. X. [8]. There are five technologies that are kept under consideration by these investigative workers and all these technologies play a key role in the vehicle to grid charging, smart charging, and the photovoltaic panels for charging of electrical vehicles, on-road charging, contactless charging etc. For electrical vehicles, there will be higher scope in the coming times and years. For using these electrical vehicles, there will be huge scope in the future and upcoming times. Since electrical vehicles take more time to charge, there is a main problem in

using these vehicles. By the use of intelligence in the fast-charging system of cars, we can solve this problem.

II. WORKING METHODOLOGY

There are many inductions of Li particle batteries found in current electric cars, a large number of these are ongoing work. Analysts can choose the electrolyte, positive and negative cathodes that try to improve exhibition, safety, and life battery range. A correlation of current Li-particle batteries used in electric cars. The types of batteries that have the largest shaded regions are considered ideal. These batteries contain the elements such as Nickel, Manganese, and Cobalt oxide and are abbreviated as (NMC), in the same way, other batteries contain Lithium, Iron, Phosphate and are abbreviated as (LFP), elements such as Lithium, manganese oxide and are abbreviated as (LMO) and as per research workers, all these batteries have best general features [9]. To upgrade the productivity of charge converters, the negative pulse charging technology was created originally for the batteries that are lead corrosive and they have been replaced with batteries containing lithium particle force that is released during the fast-charging resting periods in the batteries. Inside the cell, the negative drive reduces the voltages and limits the increase in temperature of the cells. Since a modest amount of energy is drawn from the battery by a negative heartbeat, arrangements are circulated to recover the formulated energy in the batteries. When the cells are depolarized rarely, into the batteries, there is a chance of high currents being sucked and this offers charging speed at high power and the time of charging is lowered as well. This is a way to assist the battery's synthetic answers to work totally on the battery's existence. [10]

With the similar capacity, as a matter of fact, the modern-day cells and batteries are becoming smaller and lighter in weight and this is the reason why devices with electronic powers are performing in a nicer and better way. It is always needed to acknowledge and identify that charging should be done at a slow pace for the maximum functioning of battery pack life, too far away reception of electric cars that convey natural benefits with virtually identical comfort to the current-burning based motors, charging should take place at higher power speeds to reduce the time of charging. A faster powerful excursion speed is an indication of higher charging speed, but specialized constraints are always there on each of the progress for the process of charging in modern-day electric vehicles as mentioned below:

- At the time of charging, batteries heat up and this is due to the various variables also inward obstruction and also the higher temperature can result in any injury.
- Maximum power is there in the cables and connectors that are supposed to be handled and this is estimated to be 3kW. [11].

2.1 System design and development

In this research paper, a special system has been proposed named Intelligence in Fast Charging System as shown in Figure 1) Block diagram of the proposed system. The input power to the EV charger is an alternating voltage in the range

of 230V to 420V. The EV charger uses a full-bridge resonant converter design, due to its high-power density and high switching frequency properties, to obtain direct current to charge the battery.

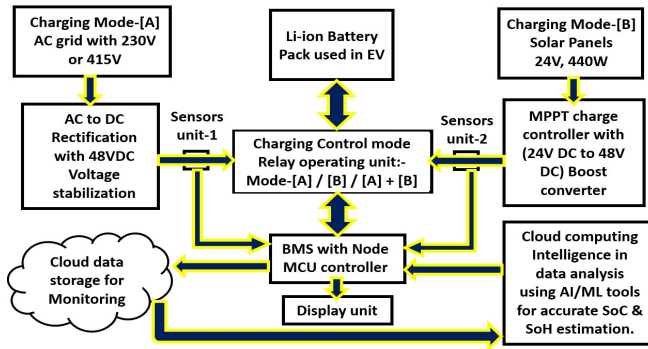


Fig. 1. Block Diagram of Proposed System.

The rectifier circuit is used by the design for the conversion of input AC voltage in the range from 230v to 420v into high 48v DC output, EMI filters are used as well for the elimination of noise of higher frequency from the source of input power. The PWM (pulse width modulation) controller can be utilized as well for MOSFETs powering the LCC converter of half bridges. With the advanced intelligence system used in BMS, the battery charging process can be evaluated and monitored. The battery voltage is monitored with the help of it along with the level of charging current and to the PWM controller IC, it offers the feedback as well. On the basis of this feedback, the duty cycle of its PWM signal is varied by the PWM controller and the MOSFET circuit is derived, and this helps to get the changing or variable voltage output for the battery charging. The relay is used for better protection for the isolation of it from the remaining part of the circuit with higher voltage compartments and this is done with the help of an opt coupler. To know the current status, the battery LED indicators are available as well. [12]

2.2 Description hardware set up

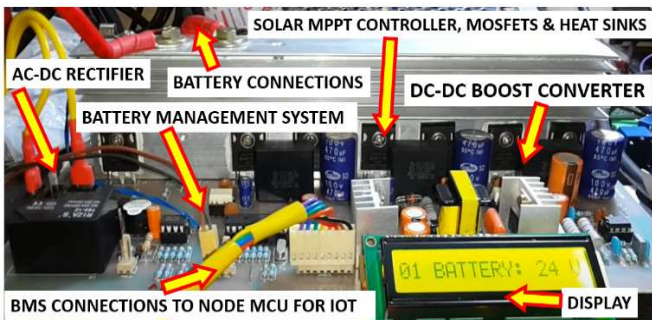


Fig. 2. Integrated Hardware of charging system with intelligent BMS.

As refer to Fig. 2) which shows the Intelligent coupled charging systems Hardware design used in the system of fast charging, with various parts of the system and their connections are shown in the diagram mentioned above. The creation of appropriate rules is needed in certain types of standards, and these are likely to address the electrical vehicles' charging behaviors. For electric vehicles, by the

simple analysis of the user behavior, we can create some simpler rules for the electric vehicles charging behaviors and their proper scheduling. We can make use of Artificial intelligence to group and categorize user behavior and predict user charging behavior. In detail, we gathered user information and used artificial intelligence to interpret the data in order to enhance the smart grid's ability to handle charging loads. By using the user's charging data on the smart grid, EV users can be categorized and the more suitable users for load balancing can be selected. Maintaining the charging and discharging behavior of EVs, can be highly significant with the help of analysis by 5G cloud networking and achievement of efficient load balancing and other modes of energy transfer that are considered reliable. [13]. On the other hand, with special charging behaviors for a small number of users, good results are not obtained by only using the methods of artificial intelligence and there is always a risk of error in the system overall. Hence, it is always suggested to make use of HITL ways as this method is considered significant for evaluating the charging behavior of the user.

III. SYSTEM ARCHITECTURE WORKING:

The designers need to be familiar with the common blocks and features mentioned in the given fig.1 for the designing of intelligence in the systems of fast charging. This is particularly important in the designing and implementation of specific circuits. These blocking features are essential in all quick chargers. Required direct current for charging at constant voltage is provided by the combined operation of charging mode-A, charging Mode-B or both Mode-A + Mode-B shown in Fig 1. The current and voltage of the battery cells are controlled by Regulators though the active charge control cell balancing technique of BMS. For the chargers that are cheaper, the power transistors or the control units are present with other elements that are compatible linearly with the power dissipation. The power is dissipated in the form of heat. For average efficiency and stepping down of the switching power supply, it can be a standard freewheel diode for maximum efficiency with a synchronous rectifier [14]. In Figure 1, the blocks mentioned on the right indicate various controlling functions and calculations. The maximum current supplied to the battery is kept limited by an analog current control loop. This is also done by a voltage loop and on the cell, a constant voltage is maintained. It is important to note that Lithium-positive cells need precision at more advanced and higher levels in the charging voltage applied. For the designing of the control system of this kind NODE MCU 8266 Micro-controller is used. This control system has the potential to turn on charging automatically at the time of discharging of the battery below its specific set point and at the time when the battery is charged to its maximum, the charger turns off. To step down transformer, the first supply is given. From the rectifier circuit afterward, the diode and capacitors are used, and they have the potential to convert alternating current power to direct current power. Whereas secondary DC power is given by 24V,440W solar panels through MPPT to DC-DC boost converter in a controlled manner though the main controller acting as intelligent synchronous BMS with two chargers are created in this way.

With the help of a relay, the charging circuit is connected to the batteries. To connect and disconnect the charger to the battery, the relay acts as a switch. NO and NC are two terminals of the relay. NO stands for normally open and NC stands for normally closed terminals. [15]. At the time of charging, the relay is normally closed. The designers make use of two relays. The relays are connected to the positive and negative terminals of the battery in such a way that one of them is connected to the positive terminal of the 1st battery and the second terminal is connected to the negative terminal of 2nd battery. For the current measurement, the CT relay is used. For control purpose the Arduino UNO is used. To Arduino, CT, PT, display, and relays are connected. Arduino works as per the output of CT and PT. The relay will close both chargers at the time when the battery needs to be charged and, in this way, charging of the battery starts. The current and voltage is measured by CT and PT and the scoreboard is used to display their values. [16]. High-quality chargers and fast chargers are used by CT and IT, hence batteries from both sides can be charged with the help at a time. I-V characteristics or the current-voltage features of the charger are programmable completely and with the help of voltage-limited current, it can be only programmed. We make the measurements for the cell temperature; termination of charges is based often on the gradient level of these evaluations. The time of charging is measured by the chargers for the intelligence block calculations. On the LCD display helping in monitoring, the current, state of charge and voltage are displayed. [17]. The working of the system is shown in this way.

3.1 Description of sub components in system:

The actual hardware prototype image is shown in Figure 3. This prototype is implemented to suggest the outcomes in actually via the hardware's practical testing. The control circuits are used by these chargers to charge the batteries quickly without having any damaging impact on cells of the battery. There are multiple parts into which we can divide the battery and control circuits can be created as well with the help of these switches that are used during battery charging process. At present, for faster recharge, there are three most dominant standards that are used, and these are illustrated as CHAdEMO, CCS and GB/T [18].

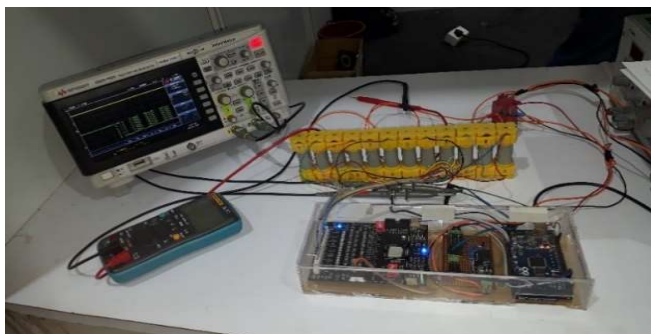


Fig. 3. Hardware Testing & Experimental set up for BMS cell balancing Operation through Node MCU.

For DC chargers, while keeping the general requirements, we can make use of a single converter for the implementation of alternating and direct current. Major challenges are involved in the stages of DC/DC for the DC charger, these challenges include high current and high voltage availability, magnetic cores with higher flux transistors, cooling systems and their specific designs. Via various sources of voltages and sometimes by the alterations of connections in two in-series current sources that are completely independent of each other. To test and determine the voltage output in parallel modules, an independent source voltage is a kind of behavior which only one of them displays. At this time, we can charge the batteries independently and separately. It is also found that the charging system at the time of charging can be made double as compared to the system of normal battery timings and with the help of such features, the battery charging time is reduced [19].

For above-mentioned hardware system, the following components are used, and these are shown below in the following figure. 2 & Figure 3: -

Solar panel with MPPT charge controller: This circuit having combined sets of MOSFETs placed on heat sinks acts as switching devices for boost controller operation in a secondary charging system connected to operate in synchronous way with AC-DC rectifier unit though IOT based Node MCU acts as BMS (Battery Management System).

The Lithium Ion Cells: From traditional and other chemical charging methods especially for the batteries with lead acid, Li-ion battery charging is totally different. To ensure maximum storage of energy in a safer manner the top charges are used. To an accuracy of 0.75 percent, the lithium charger adjusts the charging voltage and with the current limits the maximum charging speed is set as with the power supplying bench. At the time of fast charging, when it starts, there will be a decline in the cell voltage and the current reaches the current limit. At the time of charging, the battery voltage will slowly increase. In the end, there will be decline in current and as a result of it; the voltage will increase to the voltage level of breakdown per cell of 4.2V. At the time when the battery reaches, its breakdown voltage, the charger may stop charging but the charging is ignored via this approach. To start the timer when the breakdown voltage is obtained is one of the variations and after a certain period of delay end of the charging. We have another option as well and this is to keep a check on charging current and lower-level exit that is generally 5 percent of the limit. It is often suggested by some of the manufacturers to keep it to the highest of 100 mA.

Relay: A switch is called a relay and its purpose is to either connect or disconnect the circuits and the relay works automatically instead of a manual operation.

Current Sensor: The current sensor is a device to detect the signals and send these signals to the controller and this happens when the battery is ready to discharge.

Transformer: There are 230 volts in the primary coil while 120 volts are there in the secondary coil. Two packs of transformers are used by it. 230V and 5 amperes are the specifications of this transformer.

Display with LCD: LCD displays are used by this system for the electrical parameters and thermal parameters visual indication. Current, Voltage and battery state are among the electrical parameters. The charges in state and state charge are examples of it.

ESP8266 Node MCU: on the basis of higher performing IOT-based open source microcontroller works by combining the 5V power supply.

The regulator of voltage: offers +5 voltage of power to the complete circuit.

The resistors: These also help in the smoothening of the signal level for voltage division. For current limiting and measurement purposes, the system makes use of resistors of different types.

Capacitor: stabilizing voltage & filter to remove noise.

Transistor: the transistors are collectors with three terminal base emitters and act as a semiconductor device. Amplification is the main aim of these transistors. In relay circuit, the system makes use of these transistors.

Crystalline Oscillator: to generate frequency and PWM signals, the crystalline oscillators are used, and these signals are needed for the control of system overall.

IV. EXPERIMENTAL OUTCOME

Many tests were conducted after the successful prototype and it was seen that Many iterations in both the charging and battery discharging process overall, and with the intake of the mean from Figure-4) Cloud captured results in the form of graphs and its recorded values, the normal batteries take as much as 1.5 hours to get fully charged. Fast charging with the help of BMS features in a period of thirty-eight minutes and this is exactly half of the time. In this way, we can say that the system of fast charging is highly significant, and it has the potential to charge multiple batteries in a speedier way and less time is consumed. [20]. In the inverter batteries, in UPS (uninterrupted Power Supply and other batteries of electric vehicles, this system can be used efficiently. In the system of fast charging, with the help of intelligence, the battery voltage is offered by it along with the charge of the battery at the same time. With the help of an intelligence system, we can charge multiple batteries at a time and in this way time of charging is reduced and saved as well [21].

From the results mentioned above, we can draw a conclusion that when the lithium-ion battery charging time is shortened, it will allow us to run faster applications of electrical vehicles in such a way that temperature is not affected. For the system of fast charging, since the intelligence methods suggested are energy efficient methods in this paper and with the decline of higher current, there is always a reduction in the battery charging time especially for the lithium-ion batteries and in a few

cases, it must maintain its the temperature to 35 degrees centigrade for thermal optimization. This process allows the lithium-ion batteries to work efficiently and there is no effect of this optimization on the life of these batteries. [22]

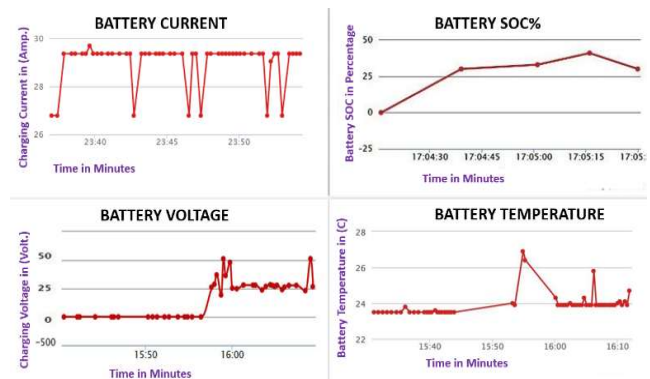


Fig. 4. Battery parameters monitoring over cloud server while charging though this proposed intelligent charging system.

V. FUTURE SCOPE

In the coming times, the world will say goodbye to conventional vehicles and only electrical vehicles will remain in use. These vehicles with the help of fast chargers can deliver enough power over a ten-minute charging cycle, that would be sufficient enough to run the vehicle for more than a hundred miles. The psychology of a wide range of fleet drivers is impacted by the fast chargers of electric vehicles, a wide adoption of electric vehicles will be allowed by comprehensive electric vehicle fast charging infrastructure. [23]. The new era of electric vehicles will be introduced by the system of fast charging and intelligence. For the system suggested in this research paper, the following are the main advantages:

- Charging time is saved by the intelligence in the system of fast charging
- Visual battery charging status is available with the help of LCD display.
- The battery is kept safer from overheating with the help of a battery indicator and the battery performance increases due to it.
- Smart IOT-based cloud data storage allows a wide scale of data collection, long-time recording & analysis in accurate levels helps in precision SoC & SoH calculations required for range estimation in electric vehicles.

It is a confirmed matter that electric vehicles are becoming essential in the coming times and future since they provide many beneficial aspects to us. Some of these beneficial aspects include no cost of fuels, no utilization of non-renewable energy sources, zero emissions, lower costs of running, highly economical and eco-friendly etc. On the other hand, the time to charge the battery is the only concern regarding the use of electric vehicles. [24]. We have learned that this concern can be avoided using the system suggested

in this investigative paper. In a battery charging system, if artificial intelligence is used, the time to charge the battery is reduced considerably. Some highly time-saving aspects regarding the mobility of vehicles are discussed in this research paper and these suggestions can be highly helpful in the future and more research is needed in this regard. The research mentioned here is not only useful for the reduction of battery timings but there are certain other advanced features mentioned in this research paper that can be highly significant for the research to be conducted in the future. The thermal stability of battery and the performance of battery are some other challenges that automobile manufacturers will face while creating electric vehicles. [25].

CONCLUSION

The use of advanced cloud computing intelligence-based BMS is suggested in this research paper for electric vehicles in the mode of faster and speedier novel charging techniques. In any type of battery charging system, this type of technology can be used in the future. Therefore, the outcomes of these experiments suggested here confirm that there is a reduction of time when fast charging is used and the overall life of the battery is improved as well. The thermal stability of the battery is even not affected by the use of fast charging. In this research paper, we have discussed the beneficial aspects of creating Intelligent BMS for integrated solar & AC grid-coupled fast charging systems for electrical vehicles and this has no impact on the thermal stability of the batteries and their life span. In designing of existing system of battery management special features are upgraded so that all safety parameters are checked and supported well. The time of charging is reduced by the use of these energy-optimizing techniques and overheating is avoided as well. When LCD is used on the battery indicators, the batteries are kept safe from overheating, and this is done by the alternation of time by BMS controlled through switch mode charging intelligence the overall impact is the increase in the performance of the battery and helps in saving the battery's life span. For this technology, the future version could bring a tremendous revolution in the EV industry, and it can overcome all the existing concerns regarding the use of electric vehicles.

REFERENCES

- [1] Du, J.; Sun, Y.; Huang, Y.; Wu, X. Analysis of Influencing Factors of Thermal Management System for LiFePO₄ Lithium Battery under High Power Charging. *World Electr. Veh. J.* 2020, 11, 44. [CrossRef]
- [2] Wolbertus, R.; Hoed, R.V.D. Electric Vehicle Fast Charging Needs in Cities and along Corridors. *World Electr. Veh. J.* 2019, 10, 45. [CrossRef]
- [3] Ciprian Ionescu, Andrei Drumea, Paul Svasta, Alexandru Vasile, Norocel Codreanu, "Evaluation of Active Balancing Circuits for Supercapacitors", Presented by C. Ionescu at the 2nd PCNS 10-13th September 2019, Bucharest, Romania as paper 5.3
- [4] Van Vliet, O.; Van Vliet, O.; Aplyn, D. The potential of electric trucks—An international commodity-level analysis. *Appl. Energy* 2019, 236, 804–814. [CrossRef]
- [5] W. Li, L. Fan, Y. Zhao, X. He, D. Xu, and B. Wu, "High Step-Up and High Efficiency FuelCell Power Generation System with Active Clamp Flyback-Forward Converter," *Industrial Electronics, IEEE Transactions on* vol. PP, no. 99, pp. 1, 2011.
- [6] K. Hatua, S. Dutta, A. Tripathi, S. Baek, G. Karimi and S. Bhattacharya, "Transformer less Intelligent Power Substation design with 15kV SiC

- IGBT for grid interconnection," in *Proc. Of IEEE Energy Conversion Congress and Exposition (ECCE)*, Phoenix, AZ, 2011, pp. 4225–4232.
- [7] D. Li, Bo Liu, Bo Yuan, Xu Yang, J. Duan, and J. Zhai, "A high step-up current fed multiresonant converter with output voltage doubler," in *Proc. 2011 IEEE Applied Power Electronics Conference and Exposition* pp. 2020–2026.
- [8] Zhang, C.; Jiang, J.; Gao, Y.; Zhang, W.; Liu, Q.; Hu, X. Charging optimization in lithium-ion batteries based on temperature rise and charge time. *Appl. Energy* 2017, 194, 569–577. [CrossRef]
- [9] Burnham, A.; Dufek, E.J.; Stephens, T.; Francfort, J.; Michelbacher, C.; Carlson, R.B.; Zhang, J.; Vijayagopal, R.; Dias, F.; Mohanpurkar, M.; et al. Enabling fast charging—Infrastructure and economic considerations. *J. Power Sources* 2017, 367, 237–249. [CrossRef]
- [10] Anseán, D.; García, V.; Gonzalez, M.; Viera, J.C.; Blanco, C.; Antuña, J. DC internal resistance during charge: Analysis and study on LiFePO₄ batteries. In *Proceedings of the 2013 World Electric Vehicle Symposium and Exhibition (EVS27)*, Barcelona, Spain, 17–20 November 2013; pp. 1–11. [CrossRef]
- [11] Wu, B.; Li, Z.; Zhang, J.; Huang, J.; Nie, Z.; Sun, Y.; An, F.; Wu, N. Thermal modelling of large-format laminated Li-ion battery and experimental validation using embedded thermocouples. In *Proceedings of the 2013 World Electric Vehicle Symposium and Exhibition (EVS27)*, Barcelona, Spain, 17–20 November 2013; pp. 1–9.
- [12] Idaho National Laboratory. Plugged in: How Americans Charge Their Electric Vehicles; Idaho National Laboratory: Idaho Falls, ID, USA, 2015; pp. 1–24.
- [13] Khoo, Y.B.; Wang, C.-H.; Paevere, P.; Higgins, A. Statistical modeling of Electric Vehicle electricity consumption in the Victorian EV Trial, Australia. *Transp. Res. Part D Transp. Environ.* 2014, 32, 263–277. [CrossRef]
- [14] Farzad Ahmadkhanlou, Abas Goodarzi and Don King, "Plug in hybrid electric vehicle with dual battery system" EVS-25 Shenzhen, China, Nov. 5-9-2010, The 25th World Battery, Hybrid and Fuel Cell Electric Vehicle Symposium & Exhibition.
- [15] Wolbertus, R.; Hoed, R.V.D.; Maase, S. Benchmarking Charging Infrastructure Utilization. *World Electr. Veh. J.* 2016, 8, 754–771. [CrossRef]
- [16] Open access peer-reviewed chapter "Batteries and Supercapacitors for Electric Vehicles" By Monzer AI Sakka, Hamid Gualous, Noshin Omar and Joeri Van Mierlo, Published: Dec. 19th, 2012 DOI: 10.5772/53490
- [17] Asamer, J.; Reinthaler, M.; Ruthmair, M.; Straub, M.; Puchinger, J. Optimizing charging station locations for urban taxi providers. *Transp. Res. Part A Policy Pr.* 2016, 85, 233–246. [CrossRef]
- [18] Schroeder, A.; Traber, T. The economics of fast charging infrastructure for electric vehicles. *Energy Policy* 2012, 43, 136–144. [CrossRef]
- [19] Sun, X.-H.; Yamamoto, T.; Morikawa, T. Fast-charging station choice behavior among battery electric vehicle users. *Transp. Res. Part D Transp. Environ.* 2016, 46, 26–39. [CrossRef]
- [20] Motoaki, Y.; Shirk, M.G. Consumer behavioral adaption in EV fast charging through pricing. *Energy Policy* 2017, 108, 178–183. [CrossRef]
- [21] K. Morrow, D. Karner and J. Francfort, "Plug-in hybrid electric vehicle charging infrastructure review" in, Washington, DC: U.S. Dept. Energy—Veh. Technol. Program, 2008
- [22] Morrissey, P.; Weldon, P.; O'Mahony, M. Future standard and fast charging infrastructure planning: An analysis of electric vehicle charging behaviour. *Energy Policy* 2016, 89, 257–270. [CrossRef]
- [23] Wolbertus, R.; Kroesen, M.; Hoed, R.V.D.; Chorus, C. Fully charged: An empirical study into the factors that influence connection times at EV-charging stations. *Energy Policy* 2018, 123, 1–7. [CrossRef]
- [24] Franke, T.; Krems, J.F. Understanding charging behaviour of electric vehicle users. *Transp. Res. Part F Traffic Psychol. Behav.* 2013, 21, 75–89. [CrossRef]
- [25] Jiao, Z.; Ran, L.; Chen, J.; Meng, H.; Li, C. Data-Driven Approach to Operation and Location Considering Range Anxiety of One-Way Electric Vehicles Sharing System. *Energy Procedia* 2017, 105, 2287–2294. [CrossRef]